

Title: Numerical Methods for Engineering: Root Finding

Objective:

To implement the Newton-Raphson iterative method in Python to solve a complex polynomial equation $f(x) = x^3 - x - 2$, demonstrating the application of numerical analysis to solve engineering problems that lack simple analytical solutions.

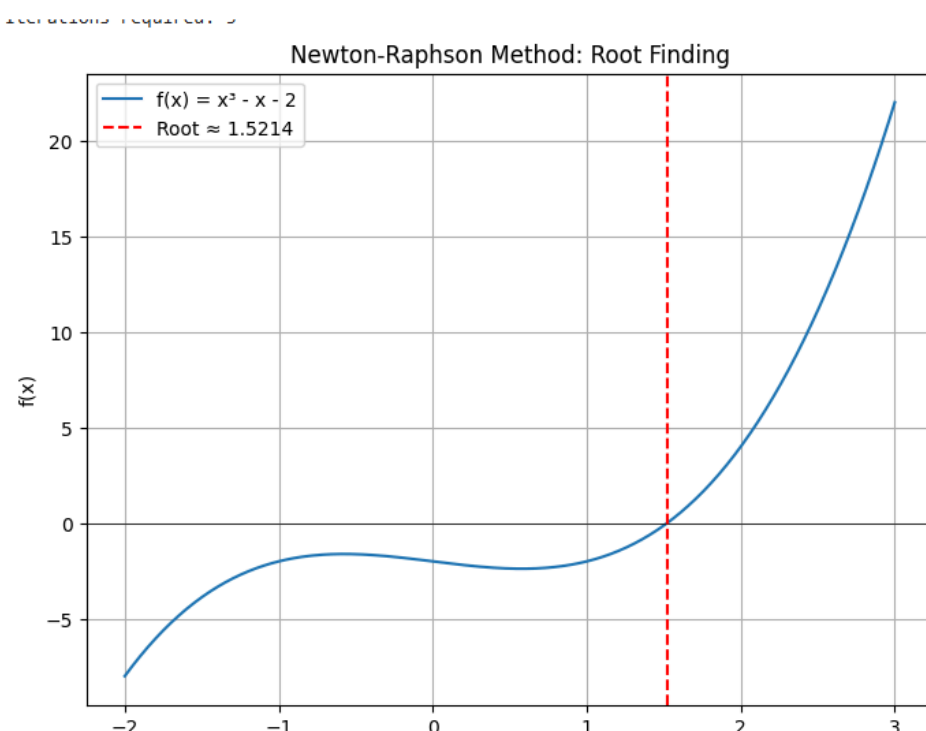
Methodology:

A Python script was developed using **NumPy** for mathematical computation and **Matplotlib** for visualization. The solver iterates using the formula $x_{n+1} = x_n - f(x_n) / f'(x_n)$ until the difference between iterations is less than a defined tolerance of 1×10^{-6} .

Results:

The algorithm successfully converged to a root of approximately 1.52138 within 4 iterations. A plot of the function was generated to visually confirm the intersection with the x-axis.

Newton-Raphson method



Conclusion & Key Learning:

This project demonstrates proficiency in translating mathematical theories into functional Python code. The ability to computationally solve complex equations is directly applicable to mechatronics, such as calculating kinematic paths, thermal models, or dynamic system responses that cannot be solved by hand.